

Cryptanalysis of Poseidon-Based Fiat-Shamir Protocols

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Abstract

Poseidon[GKR⁺19] and its successor Poseidon2[GKS23] are arithmetization-oriented permutations proposed for hashing inside algebraic proof systems. While there is extensive prior work on cryptanalysis of reduced-round Poseidon/Poseidon2 and theoretical and practical weaknesses of the Fiat-Shamir transform, these two attacks are typically studied in isolation. This work asks whether *combining* them i.e., instantiating Fiat-Shamir with Poseidon can enable new attacks.

We design and implement multiple candidate attacks that encode verifier checks together with reduced-round Poseidon-based challenge derivation as explicit polynomial systems, targeting settings such as Sigma protocols and FRI-like degree enforcement. We evaluate these approaches using Gröbner bases. We use this model to derive Gröbner-basis attack formulations aimed at constructing verifier-accepting transcripts that violate the intended low-degree bound, and we report the computational bottlenecks encountered in representative parameter regimes.

We also document additional explored directions, including retry-style challenge biasing and reduced-round investigations, in order to clarify which attacks appear more or less promising. Across our implementations, we do not obtain meaningful forgeries: even with heavily reduced Poseidon parameters, Gröbner-basis computations do not find solutions for random instances of the constructed systems. Our results provide negative evidence for several natural algebraic attack routes on Poseidon-based Fiat-Shamir instantiations, and help clarify which known Fiat-Shamir weaknesses do and do not transfer to well-structured, fully transcript-bound deployments.

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1 Introduction

Poseidon and Poseidon2 are cryptographic hash functions specifically designed for efficient use in zero-knowledge proof systems. Unlike traditional hash functions optimized for bit-level operations, Poseidon-style constructions operate natively over finite fields and minimize the number of arithmetic constraints required in zk-circuits. As a result, they are becoming a standard choice in modern SNARKs and STARKs. The Fiat–Shamir transform is a fundamental technique for converting interactive proof protocols into non-interactive ones by replacing verifier randomness with the output of a hash function(which can be Poseidon or Poseidon2) applied to the protocol transcript. This transform is widely used in zk-proofs and is known to be secure in the Random Oracle Model. There exists a substantial body of work analyzing algebraic and statistical attacks on reduced-round Poseidon and Poseidon2, as well as a separate line of research examining theoretical and practical weaknesses of the Fiat–Shamir transform. However, currently there is no systematic analysis that combines these two attacks. In particular, little is known about whether the algebraic structure exploited in Poseidon cryptanalysis can interact with Fiat–Shamir–based challenge generation to produce new protocol-level vulnerabilities.

1.1 Our Contribution

This project focused on assessing whether known and emerging cryptanalytic ideas against Poseidon-style permutations, and Fiat-Shamir transform can be combined to turn into practical attack against Poseidon-instantiated Fiat–Shamir proof systems. Our contribution is twofold:

- A structured research of the attacks against reduced-round Poseidon instances, and against Fiat–Shamir transformation.
- Multiple implemented attack prototypes, including:
 1. Algebraic-solving attempts (Gröbner basis) on Sigma and FRI-based protocols with Poseidon-based FS.
 2. Contrived ROM-secure but instantiation-insecure Schnorr-style protocol
 3. Extension of the GKR attack concept using higher-degree “intersection” constructions

1.2 Related Work

Poseidon is built around a HADES-style permutation with full and partial rounds. Algebraic cryptanalysis of HADES-based designs has been studied in multiple works, including systematic analyses of the design strategy and its implications for algebraic attacks.

1.2.1 Algebraic attacks on reduced-round Poseidon

Modern attacks exploit this structure using algebraic tools such as: polynomial interpolation, resultants, Gröbner bases and degree-growth analysis. Recent works explore advanced polynomial-system techniques for arithmetization-oriented primitives, including Graeffe-based root-finding methods, subspace-trail exploitation, and improved elimination strategies.

1.2.2 Weak Fiat–Shamir attacks

A well-understood class of failures arises when Fiat–Shamir is applied with an incomplete transcript, most commonly when public parameters are not hashed into the challenge. In this weak FS setting, an adversary can create a valid proof, but effectively for a different statement than intended (e.g., proving knowledge of some witness rather than the intended one). This issue has appeared in many real implementations. In our work, this line is largely out of scope as we focus on well-structured implementations that include the full transcript, which removes the necessity to exploit weak FS attacks.

1.2.3 Practical and theoretical Fiat–Shamir attacks

A recent work by [KRS25] demonstrates concrete breakage of the GKR[GKR08] protocol when Fiat–Shamir is initiated with any specific hash function. A large body of theory shows that even if Fiat–Shamir is secure in the Random Oracle Model, this does not automatically imply security under any concrete hash function instantiation. These works construct contrived protocols (protocols which are designed to fail) that remain secure in the ROM but become insecure once the random oracle is replaced by a fixed hash function.

2 Background and Notation

2.1 Fields, rings, and vectors

We work over a prime field $\mathbb{F} = \mathbb{F}_p$ for a prime p . All arithmetic operations are in $\mathbb{F}$ unless stated otherwise. Vectors are denoted in bold, e.g., $\mathbf{x} = (x_1, \dots, x_n)$. The multivariate polynomial ring in variables $\mathbf{z} = (z_1, \dots, z_N)$ is $\mathbb{F}[\mathbf{z}] = \mathbb{F}[z_1, \dots, z_N]$. Univariate polynomials are written in $\mathbb{F}[X]$.

2.2 Poseidon

Poseidon[GKR⁺19] was introduced as a field-native hash function optimized for arithmetization, designed around a permutation-based sponge construction. Poseidon-family hashes are defined from a permutation $\pi : \mathbb{F}^t \rightarrow \mathbb{F}^t$ and then used in sponge or compression mode. The permutation acts on a width- t state, and the hash interface specifies how inputs are absorbed, how many permutation calls are performed, and how outputs are squeezed or extracted. Poseidon-family designs follow the HADES[ABM23] strategy: the permutation uses R_f full rounds, where the S-box is applied to all state elements, and R_p partial rounds, where the S-box is applied to a strict subset of elements (often one). Intuitively, full rounds provide strong global mixing while partial rounds reduce multiplicative cost, aiming to maintain diffusion properties at lower arithmetic complexity. For algebraic cryptanalysis, the central point is that a permutation call can be represented as a set of low-degree polynomial constraints linking the input state,

intermediate round states, and output state. In prime-field Poseidon instances, the nonlinearity is an S-box $x \mapsto x^\alpha$ with small odd α (often $\alpha = 5$), yielding low-degree equations. From a modeling viewpoint, each round is typically decomposed as:

1. **AddRoundConstants:** affine update $\mathbf{s} \leftarrow \mathbf{s} + \mathbf{c}_r$.
2. **S-box layer:** nonlinear map on selected coordinates.
3. **Linear diffusion:** multiplication by a matrix or application of a structured linear map.

2.2.1 Poseidon2

Poseidon2[GKS23] maintains the same high-level permutation skeleton and typically the same S-box choice for prime fields, but modifies the diffusion layers to reduce the arithmetic cost. For algebraic modeling, this implies that:

- the nonlinear constraints remain of the same qualitative form as Poseidon (same exponent map),
- the linear constraints change according to the new diffusion structure,
- the total system remains low-degree but may have different sparsity and elimination behavior.

2.2.2 Poseidon2b

Poseidon2b[GKK⁺25] is designed for binary extension fields. Although finite-field arithmetic and low-degree structure remain, the field characteristic changes the behavior of certain exponentiation maps and may change practical arithmetization costs and cryptanalytic heuristics. In this work, the primary relevance is methodological: the same approach can be transferred to Poseidon2b by changing the base field and the parameter set.

2.3 Fiat-Shamir

Many ZK-proof systems start with an interactive public-coin protocol, where a verifier sends a random challenge after seeing the prover’s commitment. The Fiat–Shamir (FS) transform removes interaction by replacing that verifier challenge with a hash of the transcript. If the hash behaves like a truly random function, then the prover cannot predict the challenge before fixing the commitment. Intuitively, FS preserves soundness because it restores the “commit-then-challenge” structure: the challenge is determined by the commitment and all the public context, rather than being chosen by the prover.

3 Attack Vectors

This section describes concrete attack vectors that we explored against Poseidon-based Fiat-Shamir protocols. Most of the results here are negative. Still, documenting these attempts is useful: it clarifies which directions look promising at first sight, how they can be formalized, and why they appear to fail in practice.

3.1 FRI Gröbner Basis Attack

This attack explores whether STARK[BSBHR18]/FRI-based proof systems can be broken by translating their soundness conditions into explicit algebraic constraints and solving the resulting system using Gröbner basis techniques. Gröbner bases are a powerful tool in computational algebra that transform systems of polynomial equations into a canonical form, making them easier to solve. In cryptanalysis, they have been successfully applied to break various cryptographic

primitives by exploiting their algebraic structure. The key idea is to model the entire verification logic—including the hash function, commitment scheme, and protocol constraints—as a single polynomial system, then attempt to find a solution that would represent a valid forgery.

3.1.1 Setting and Intuition

We work over a prime field $\mathbb{F}_p$. Consider k (univariate) polynomials

$$f_1(X), \dots, f_k(X) \in \mathbb{F}_p[X],$$

where each f_j may have degree up to $N + d$ for some $d > 0$, while the protocol aims to enforce a degree bound $< N$. The prover commits to evaluations on a public evaluation domain

$$D = \{a_1, \dots, a_m\} \subset \mathbb{F}_p, \quad \ell_{j,t} = f_j(a_t) \quad (t = 1, \dots, m).$$

Each list of evaluations is committed using a Merkle tree (with Poseidon as the compression hash), producing roots $\text{MerkleRoot}_1, \dots, \text{MerkleRoot}_k$.

To enforce the degree bound, the verifier checks a *random linear combination* of these polynomials. In the FS version, the randomness is generated deterministically from the transcript via Poseidon:

$$c = \text{Poseidon}(\text{ctx}, \text{MerkleRoot}_1, \dots, \text{MerkleRoot}_k),$$

where ctx denotes all public parameters and domain separators included in the hash input. From c , the protocol derives combination weights $\alpha_1, \dots, \alpha_k \in \mathbb{F}_p$ (for instance by taking powers $\alpha_j = c^j$, or by expanding c into multiple field elements). The combined polynomial is

$$Q(X) = \sum_{j=1}^k \alpha_j f_j(X),$$

and the low-degree check requires $\deg(Q) < N$.

The main observation behind the attack is that $\deg(Q) < N$ is equivalent to requiring that *all coefficients of degrees N through $N + d$ vanish* in Q . This creates explicit algebraic constraints that interact with the FS challenge and the Merkle commitment.

3.1.2 Algebraic System

We represent each polynomial by its coefficients:

$$f_j(X) = \sum_{i=0}^{N+d} C_{j,i} X^i, \quad C_{j,i} \in \mathbb{F}_p.$$

The high-degree coefficients are $\{C_{j,i}\}_{i=N}^{N+d}$. Writing the coefficient of X^i in Q gives

$$[X^i] Q(X) = \sum_{j=1}^k \alpha_j C_{j,i}.$$

Therefore, the degree bound $\deg(Q) < N$ yields the constraints

$$\sum_{j=1}^k \alpha_j C_{j,i} = 0 \quad \text{for all } i \in \{N, \dots, N + d\}.$$

To couple these constraints to Fiat-Shamir and commitments, we build one joint polynomial system whose unknowns include:

- polynomial coefficients $C_{j,i}$ and evaluation values $\ell_{j,t}$,
- Merkle tree internal node values and roots MerkleRoot_j ,
- Poseidon internal state variables for all hash invocations,
- the derived challenge c and the combination weights α_j .

The system contains several groups of equations:

- **Evaluation constraints:**

$$\ell_{j,t} - \sum_{i=0}^{N+d} C_{j,i} a_t^i = 0 \quad (j = 1, \dots, k; t = 1, \dots, m).$$

- **Merkle constraints:** for each internal node,

$$\text{parent} - \text{PoseidonHash}(\text{left}, \text{right}) = 0,$$

with analogous constraints for all nodes up to MerkleRoot_j .

- **Poseidon FS constraints:** round-by-round polynomial equations for the Poseidon permutation (input includes ctx and all Merkle roots; output constrained to the challenge c).
- **Degree-bound constraints:** the vanishing of high-degree coefficients of Q :

$$\sum_{j=1}^k \alpha_j C_{j,i} = 0 \quad (i = N, \dots, N + d).$$

Once this polynomial system is constructed, we apply Gröbner basis algorithms to find a solution. The Gröbner basis computation transforms the ideal generated by these equations into a triangular or reduced form, potentially revealing values for the unknowns. Standard algorithms like Buchberger’s algorithm or F4/F5 can be used, though their complexity grows rapidly with the number of variables and the degree of equations.

3.1.3 Attack Goal and Outcome

Given public transcript data (e.g., the Merkle roots and/or the FS output values), the attack fixes these as constants and tries to find a satisfying assignment for the remaining variables by computing a Gröbner basis of the generated ideal. A successful solution would correspond to a forged transcript where $\deg(Q) < N$ holds algebraically, even though the underlying polynomials f_j contain non-zero high-degree parts.

We implemented this approach by encoding the full verification logic as a polynomial system and attempting to solve it. Even when we reduced Poseidon to a much smaller number of rounds (to make the algebraic system easier), we did not obtain solutions for randomly generated instances. The results suggest that the interaction between the degree-cancellation constraints, the Merkle structure, and the FS-derived challenge produces systems that are not currently tractable for generic Gröbner basis solving.

Table 1: Groebner elimination basis results (no Poseidon trace vars). Fixed params across runs: $p = 65519$, $R_F = 2$, $R_P = 1$, $t = 3$, **rate** = 2, **num_zero_polys** = 1, $\alpha = 3$, $\beta = 5$. If a Groebner basis is found, we report its size $|\text{GB}|$.

N	m	n	k	$ \text{GB} $
1927	10	1000	3	$\nexists$
82	10	10	3	199
94	10	10	3	221
47	10	10	3	127
47	15	10	3	127
47	30	10	3	$\nexists$
47	10	20	3	$\nexists$
47	10	15	3	127
47	10	15	7	127

Benchmarks The evaluation domain is an N -point multiplicative subgroup with (chosen such that $N \mid (p - 1)$). We use m polynomials with low-degree threshold n and “excess” degree k , i.e., a forge bound $\deg < n + k$. To keep commitments algebraic, Merkle internal nodes were constrained via a linear hash

$$H(L, R) = \alpha L + \beta R,$$

and the Poseidon sponge absorbed the transcript (by default) $[N, m, \text{root}]$ and produced a challenge c enforced by $c = \text{PoseidonOutput}[0]$. We additionally included zero constraints of the form $P_j(c) = 0$ on a chosen subset of polynomials. We tested two coefficient-restriction variants:

- A weakened restriction constraining only a single linear condition on the excess-degree part, which we used for the main runtime attempt.
- Forcing all excess-degree coefficients to 0 (enforcing $\deg < n$), which did not yield a usable outcome; $N = 3854$, $m = 10$, $n = 1000$ and $k = 10$. In that configuration, the polynomial ring contained 17,828 variables; using **degrevlex** ordering on a MacBook Pro (M2 Pro), the Gröbner basis computation ran for approximately 4 hours before we terminated it, without returning a basis.

3.2 Sigma Protocol Algebraic Attack

A similar attack based on GB can be designed against sum-check or Schnorr’s sigma protocol, here we study whether a Σ -protocol becomes vulnerable when the FS challenge is instantiated with an algebraic hash function (Poseidon), by attempting to learn an accepting transcript via algebraic solving.

3.2.1 Attack Model

A typical Σ -protocol has three moves:

1. the prover sends a commitment $A = \text{Commit}(r)$ for random r ,
2. the verifier sends a challenge c ,
3. the prover answers with a response $z = \text{Resp}(x, r, c)$, often of the form $z = r + c \cdot x$.

The verifier then checks a relation $\text{Verify}(\text{statement}, A, c, z) = 1$ (e.g., for Schnorr: $g^z = A \cdot y^c$).

In the Fiat-Shamir transform, the challenge is computed as

$$c = H(\text{ctx}, \text{statement}, A),$$

and in our setting H is instantiated with Poseidon.

3.2.2 Algebraic Encoding

The attack treats the transcript elements as unknowns and aims to satisfy both:

- **Hash constraints:** $c = \text{Poseidon}(\text{ctx}, \text{statement}, A)$, expanded into round-by-round polynomial equations (including all internal Poseidon state variables).
- **Verification constraints:** the protocol verification equation, encoded in the same algebraic framework (either directly in the field, or via algebraic constraints corresponding to the group/arithmetic representation used by the protocol).

At a high level, this produces a polynomial system in variables such as (A, c, z) (and possibly r and auxiliary variables), and the attacker attempts to find a solution using Gröbner bases or related elimination techniques (e.g., resultants). If successful, such a solution would give a forged proof without knowledge of the witness.

3.2.3 Attack Goal and Outcome

We implemented this approach in Sage and tested it under very weak settings (reduced-round Poseidon and small field) to make solving as favorable as possible. Even in these simplified configurations, we did not obtain a valid solution in reasonable time. Empirically, the coupling between Poseidon constraints and the verification equation appears to create a system that is too constrained and too nonlinear for current generic algebraic solvers. These experiments suggest that a direct “solve-for-a-transcript” strategy is not an efficient, even when the hash is significantly weakened. While this does not rule out more specialized cryptanalysis, it provides evidence that naive algebraic attacks do not immediately apply in the simplest single-round case.

Benchmarks To probe the effect of transcript-derived challenges in a much smaller setting, we also benchmarked a “Schnorr-like” scalarized transcript-consistency system over the same field $\mathbb{F}_p$ with $p = 65519$ and the same reduced-round Poseidon instance ($t = 3$, rate 2, capacity 1, $(R_F = 2, R_P = 1)$) as we did with FRI. For tractability this uses an additive/scalar model rather than an elliptic-curve group: generator scalar $G = 5 \bmod p$, secret scalar $y = 7$ (used only to define the public key), and public-key scalar $Y = y \cdot G \bmod p$, with transcript constants `domain = 123` and `params = 456`. The system enforces $c = \text{PoseidonOutput}[0]$ for a sponge absorbing the transcript (including A), and includes the verification-style constraint

$$zG = A + cY \quad \text{in } \mathbb{F}_p.$$

This ring had 17 variables in total. Using Sage’s default monomial ordering, the Gröbner basis computation completed and returned a basis of size 28 polynomials. However, attempting an order change to lexicographic order ran for approximately 3 hours on the same MacBook Pro (M2 Pro) and was terminated without producing a result.

3.3 Contrived Extended Schnorr Protocol

This attack vector illustrates a conceptual gap between security in the Random Oracle Model (ROM) and security under a concrete hash instantiation. We constructed a deliberately artificial extension of Schnorr identification that remains secure in the ROM, but becomes completely insecure once the FS challenge is computed by a deterministic hash function.

3.3.1 Protocol Description

Let g be a generator of a group of prime order, and let $y = g^x$ be the public key for secret x . The protocol proceeds in rounds.

1. The prover sends a Schnorr commitment $A = g^r$, for a random r .
2. If the last bit of c is 0, the prover sends the standard response $z = r + cx$, and the verifier checks $g^z = A \cdot y^c$. The protocol terminates.
3. If the last bit of c is 1, the verifier *skips verification* and moves to round 1 with a new round and a fresh commitment.
4. The verifier repeats this for up to N rounds; if it never sees a challenge ending in 0, it accepts without performing any Schnorr verification.

3.3.2 Soundness analysis

In the ROM, challenges are uniform and unpredictable. A cheating prover who does not know x can only succeed if all N challenges end in 1, which happens with probability 2^{-N} , since each challenge independently has a 50 percent chance of ending with 1.

However, when replacing the Random Oracle with any hash function the challenge becomes:

$$c = H(\text{ctx}, A, (\text{previous transcript})).$$

Now the prover controls A , and can therefore *grind* commitments: it can try different r values until the last bit of $H(\text{ctx}, g^r, \dots)$ becomes 1. This can be repeated in each round, therefore, the malicious prover can always ensure that the verifier never reaches the verification step, causing the protocol to accept every time.

3.3.3 Attack Goal and Outcome

This attack was inspired by theoretical attacks on Fiat–Shamir, where the idea was to design a contrived protocol which breaks with every hash function though it is secure under the Random Oracle Model. While useful for the intuition as this attack shows that if a protocol includes branching logic that depends on specific patterns in the challenge, then replacing a random oracle by a deterministic hash can enable transcript manipulation through repeated commitment attempts, it does not model real-world proof systems.

4 Conclusion

This paper investigated the security of *Poseidon-based Fiat-Shamir instantiations* from the perspective of *combined attack surfaces*: (i) algebraic cryptanalysis of reduced-round Poseidon/Poseidon2 and (ii) known theoretical and practical weaknesses of the Fiat-Shamir transform. Motivated by the fact that Poseidon-family hashes are designed to be efficient in arithmetized settings, we asked whether their algebraic structure can be leveraged to obtain protocol-level breaks when used as the Fiat-Shamir hash in modern proof systems. On the hash-function side, we examined the state of the art in *reduced-round Poseidon/Poseidon2* cryptanalysis, focusing on algebraic techniques commonly used against arithmetization-oriented primitives, including polynomial interpolation, *resultant-based* elimination, *Gröbner-basis* methods, degree-growth/complexity analyses, and *Graeffe-transform*-based root-finding approaches (as well as related structural ideas such as subspace and zero-sum distinguishers). We also examined the main known classes of Fiat-Shamir weaknesses, including weak Fiat-Shamir, theoretical separations between Random Oracle Model security and concrete instantiation, and the recent

practical attack against Fiat-Shamir in the GKR setting. Guided by the common patterns in these works, we designed and analyzed several new candidate attack directions and implemented concrete instances of them. Across all implemented approaches, we did not obtain meaningful practical attacks against well-structured, fully transcript-bound instantiations. While these outcomes are negative, they provide useful evidence about which natural attack routes *do not* appear to scale. We view these results as a map of explored and eliminated directions, together with a set of lessons for future work. In particular, our study suggests that progress is more likely to come from protocol-specific structural vulnerabilities (in the spirit of the GKR attack) or hybrid strategies that aim to bias or correlate Fiat-Shamir challenges rather than solve full end-to-end polynomial systems. We hope that the constructions, implementations, and negative results reported here help narrow the search space and inform subsequent cryptanalytic efforts on Poseidon-based Fiat-Shamir protocols.

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